A FINITE TYPE MODULE FLAT OVER THE BASE AND FINITELY PRESENTED OVER A SCHEME THEORETICALLY DENSE OPEN IS FINITELY PRESENTED

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Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file flat.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. Preamble provenance: the chapter's own retained minimal preamble (sources/stacks-flat/retained-preamble.tex, lines 1-85) is a verbatim excerpt of the native Xy-pic setup and of the native theorem declarations and mathematical macros, followed by the project's own reference/slogan/history display-environment definitions copied mechanically from the shared preamble; the wrapper class and the amsmath/amssymb/amsthm/graphicx base packages are supplied by the assembler, and extra_packages is left empty because everything needed is already in the retained preamble. No macro was invented or redefined.

Editorial note. Same-chapter pointers are located at their \label lines inside the pinned chapter file; cross-chapter pointers are located in the project's own tag list staged in the snapshot (sources/stacks-chow/tags.txt), at the verbatim lines carrying their canonical tags.

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Allowed standard prerequisites (editorial).

- A quasi-compact open of an affine scheme is a finite union of principal opens. Algebra, label algebra-lemma-qc-open, invoked at proof line 2965. Writes the dense open U as a finite union of distinguished opens D(f_i), which is what makes the injectivity of A -> product of the localisations available.
- Finiteness of presentation at a point for a flat finite type module, and openness of flatness. Same chapter flat.tex, label lemma-finite-type-flat-at-point-local (proof line 2970); Algebra, label algebra-theorem-openness-flatness (proof line 2981). Gives that the localisation of the module at the chosen prime is finitely presented over the local ring and, after constructing a model that agrees with it there, that the model is flat over the base on a neighbourhood of the point.
- A local presentation argument, and the local nature of finite presentation. Algebra, labels algebra-lemma-cover (proof line 2996) and algebra-lemma-extension (used in the same section). The cover lemma lets a local finite presentation statement be concluded globally, which is how the proof finishes from the affine neighbourhood found at the chosen point.

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Source reference. Original lemma 00EO: sources/stacks-chow/tags.txt, source lines 542–542. Navigation pointer for \ref{algebra-lemma-cover} used inside the counted span (line 2996); the argument that consumes it is retained in the span.

Source reference. Original lemma 00F6: sources/stacks-chow/tags.txt, source lines 560–560. Navigation pointer for \ref{algebra-lemma-qc-open} used inside the counted span (line 2965); the argument that consumes it is retained in the span.

Source reference. Original theorem 00RC: sources/stacks-chow/tags.txt, source lines 998–998. Navigation pointer for \ref{algebra-theorem-openness-flatness} used inside the counted span (line 2981); the argument that consumes it is retained in the span.

Source reference. Original lemma 05I7: sources/stacks-flat/pinned-original-source.tex, source lines 2879–2879. Navigation pointer for \ref{lemma-finite-type-flat-at-point-local} used inside the counted span (line 2970); the argument that consumes it is retained in the span.

Lemma 0.1. *Let $f : X \to S$ be a morphism of schemes. Let $\mathcal{F}$ be a quasi-coherent $\mathcal{O}_X$-module. Let $U \subset S$ be open. Assume*

(1) *f is locally of finite presentation,*
(2) *$\mathcal{F}$ is of finite type and flat over S,*
(3) *$U \subset S$ is retrocompact and scheme theoretically dense,*
(4) *$\mathcal{F}|_{f^{-1}U}$ is of finite presentation.*

Then $\mathcal{F}$ is of finite presentation.

Proof. The problem is local on X and S, hence we may assume X and S affine. Write $S = \mathrm{Spec}(A)$ and $X = \mathrm{Spec}(B)$. Let N be a finite B-module such that $\mathcal{F}$ is the quasi-coherent sheaf associated to N. We have $U = D(f_1) \cup \ldots \cup D(f_n)$ for some $f_i \in A$, see Algebra, Lemma 00F6. As U is schematically dense the map $A \to A_{f_1} \times \ldots \times A_{f_n}$ is injective. Pick a prime $\mathfrak{q} \subset B$ lying over $\mathfrak{p} \subset A$ corresponding to $x \in X$ mapping to $s \in S$. By Lemma 05I7 the module $N_{\mathfrak{q}}$ is of finite presentation over $B_{\mathfrak{q}}$. Choose a surjection $\varphi : B^{\oplus m} \to N$ of B-modules. Choose $k_1, \ldots, k_t \in \mathrm{Ker}(\varphi)$ and set $N' = B^{\oplus m}/\sum Bk_j$. There is a canonical surjection $N' \to N$ and N is the filtered colimit of the B-modules N' constructed in this manner. Thus we see that we can choose $k_1, \ldots, k_t$ such that (a) $N'_{f_i} \cong N_{f_i}$, $i = 1, \ldots, n$ and (b) $N'_{\mathfrak{q}} \cong N_{\mathfrak{q}}$. This in particular implies that $N'_{\mathfrak{q}}$ is flat over A. By openness of flatness, see Algebra, Theorem 00RC we conclude that there exists a $g \in B$, $g \notin \mathfrak{q}$ such that N'_g is flat over A. Consider the commutative diagram

$$\begin{array}{ccc} N'_g & \longrightarrow & N_g \\ \downarrow & & \downarrow \\ \prod N'_{gf_i} & \longrightarrow & \prod N_{gf_i} \end{array}$$

The bottom arrow is an isomorphism by choice of $k_1, \ldots, k_t$. The left vertical arrow is an injective map as $A \to \prod A_{f_i}$ is injective and N'_g is flat over A. Hence the top horizontal arrow is injective, hence an isomorphism. This proves that N_g is of finite presentation over B_g. We conclude by applying Algebra, Lemma 00EO. $\square$